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2016-2017

M1 MACRO

Group B

Session 18

These notes are written during class. They are posted "for the record" and may contain mistakes. They are not intended to be read by anyone that did not attend the class.

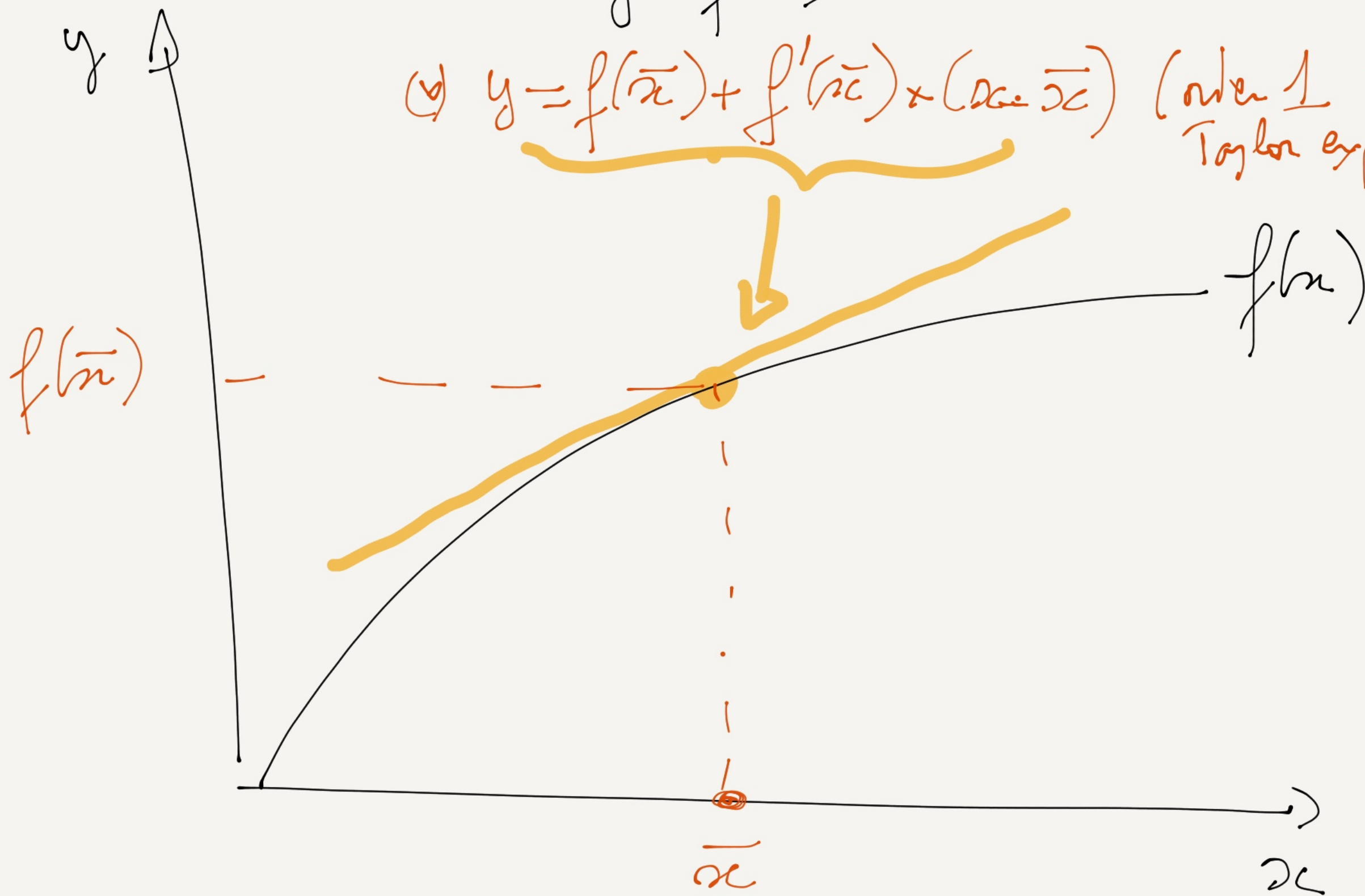
$$\ddot{X}_t - \beta b E_f X_{t+1} = \frac{1}{Z_t}$$

$$X_t = \frac{1}{C_t - b C_{t-1}}$$

linear approximation

$$y = f(x)$$

$$(v) \quad y = f(\bar{x}) + f'(\bar{x})(x - \bar{x}) \quad (\text{order 1 Taylor exp})$$



order 1 Taylor exp.

$$y_t = f(x_{1t}, x_{2t})$$

steady state $\bar{y}, \bar{x}_1, \bar{x}_2$

Taylor exp around the SS

$$\underbrace{f(x_{1t}, x_{2t})}_{\bar{y}} = \underbrace{f(\bar{x}_1, \bar{x}_2)}_{\bar{y}} + \underbrace{\bar{x}_1 f'_1(\bar{x}_1, \bar{x}_2)}_{\bar{y}} \underbrace{\left(\frac{x_{1t} - \bar{x}_1}{\bar{x}_1} \right)}_{\hat{x}_{1t}} + \underbrace{\bar{x}_2 f'_2(\bar{x}_1, \bar{x}_2)}_{\bar{y}} \underbrace{\left(\frac{x_{2t} - \bar{x}_2}{\bar{x}_2} \right)}_{\hat{x}_{2t}}$$

if $y = f(x)$

$$\frac{x f'}{y} = \varepsilon_{\text{elasticity}} \frac{f}{f'}$$

$$\bar{y} = f(\bar{x}_1, \bar{x}_2)$$

Approx : $\hat{y}_t = \frac{\bar{x}_1 f'_1}{f} \hat{x}_{1t} + \frac{\bar{x}_2 f'_2}{f} \hat{x}_{2t}$

Example

$$y_t = z_t h_t^\alpha = f(z_t, h_t)$$

$$\hat{y}_t = \frac{\bar{z} \overset{= \bar{h}^\alpha}{\underbrace{f'_1}_{\underbrace{f}_{\bar{z} h^2}}}}{\underbrace{f}_{\bar{z} h^2}} \hat{z}_t + \frac{\bar{h} \times \overset{= \bar{z} \alpha \bar{h}^{\alpha-1}}{\underbrace{f'_2}_{\underbrace{f}_{\bar{z} h^2}}}}{\underbrace{f}_{\bar{z} h^2}} \hat{h}_t$$

$$\rightarrow \boxed{\hat{y}_t = \hat{z}_t + \alpha \hat{h}_t}$$

other example

$$y_t = c_t + i_t + g_t = f(c_t, i_t, g_t)$$

$$\hat{y}_t = \frac{\bar{c} f_1'}{f_y} \hat{c}_t + \frac{\bar{i} f_2'}{f_y} \hat{i}_t + \frac{\bar{g} f_3'}{f_y} \hat{g}_t$$

$$\hookrightarrow \boxed{\hat{y}_t = \frac{\bar{c}}{f_y} \hat{c}_t + \frac{\bar{i}}{f_y} \hat{i}_t + \frac{\bar{g}}{f_y} \hat{g}_t}$$

ex. 3

$$g(y_{1t}, y_{2t}) = f(x_{1t}, x_{2t})$$

at the ss: $f = g$



$$\frac{\bar{y}_1 g'_1}{g} \bar{y}_{1t} + \frac{\bar{y}_2 g'_2}{g} \bar{y}_{2t}$$

$$= \frac{\bar{x}_1 f'_1}{f} \bar{x}_{1t} + \frac{\bar{x}_2 f'_2}{f} \bar{x}_{2t}$$

Back to the model

$$\left\{ \begin{array}{l} X_t = \beta b \sum_{\tau} X_{t-1} = \frac{1}{Z_t} \quad (1) \end{array} \right.$$

$$X_t = \frac{1}{K_t - b C_{t-1}} \quad (2)$$

SS $\bar{Z}$ given

$$(2) \Rightarrow \bar{X} = \frac{1}{(1-b)\bar{Z}}$$

$$(1) (1-\beta b)\bar{X} = \frac{1}{\bar{Z}} \Rightarrow \bar{X} = \frac{1}{(1-\beta b)\bar{Z}}$$

$$\hookrightarrow \bar{C} = \frac{(1-\beta b)\bar{Z}}{1-b}$$

$$X_t - \beta b E_t X_{t+1} = \frac{1}{Z_t}$$

$$f(X_t, X_{t+1}) = g(Z_t)$$

Linearization

$$\frac{\bar{X} \times \underbrace{f'_1}_{(1-\beta b)\bar{X}} \hat{X}_t + \frac{\bar{X} \times \underbrace{f'_2}_{(1-\beta b)\bar{X}} E_t \hat{X}_{t+1}}{\underbrace{f}_{(1-\beta b)\bar{X}}} = \frac{\bar{Z} \underbrace{g'}_{1/\bar{Z}} \hat{Z}_t}{\underbrace{g}_{1/\bar{Z}}}$$

$\nearrow -\frac{1}{\bar{Z}^2}$

$$\frac{1}{1-\beta b} \hat{X}_t - \frac{\beta b}{1-\beta b} E_t \hat{X}_{t+1} = -\hat{Z}_t$$

$\Rightarrow$

$$\hat{X}_t = \beta b E_t \hat{X}_{t+1} - (1-\beta b) \hat{Z}_t$$

Solving $\hat{X}_t = \beta b E_t \hat{X}_{t+1} - (1-\beta b) \bar{Z}_t$

forward substitution, with $0 < \beta b < 1$

$$\begin{aligned} \hat{X}_t &= -(1-\beta b) \bar{Z}_t + \beta b E_t \left[-(1-\beta b) \bar{Z}_{t+1} + \beta b E_{t+1} \hat{X}_{t+2} \right] \\ &= -(1-\beta b) \bar{Z}_t - (1-\beta b) \beta b E_t \bar{Z}_{t+1} + (\beta b)^2 E_t \hat{X}_{t+2} \\ &= \dots \text{again and again} \dots \end{aligned}$$

$$\hat{X}_t = -(1-\beta b) \underbrace{\sum_{j=0}^{\infty} (\beta b)^j E_t \bar{Z}_{t+j}}_{\text{CONVERGING}} + \underbrace{\lim_{j \rightarrow \infty} (\beta b)^j E_t \hat{X}_{t+j}}_{= 0 \text{ (TRANSIENT)}}$$

Assume: $\bar{z}_t = \rho \bar{z}_{t-1} + \varepsilon_t$

then $\varepsilon_t \bar{z}_{t+j} = \rho^j \bar{z}_t$

$0 < \rho < 1$

and therefore

$$\hat{x}_t = -(1 - \beta b) \bar{z}_t \times \sum_{j=0}^{\infty} (\beta b \rho)^j$$

Solution

$\frac{1}{1 - \beta b \rho}$

$$\hat{x}_t = \frac{-(1 - \beta b)}{1 - \beta b \rho} \bar{z}_t$$

What about $\hat{c}$?

$$X_t = \frac{1}{c_t - b c_{t-1}} = f(c_t, c_{t-1})$$

↳ linear approx: $\hat{X}_t = \frac{\bar{c} \cdot f'_1}{f} \hat{c}_t + \frac{\bar{c} f'_2}{f} \hat{c}_{t-1}$

$$f'_1 = -\frac{1}{((1-b)\bar{c})^2}$$

$$f'_2 = +\frac{b}{((1-b)\bar{c})^2}$$

$$\hat{X}_t = \frac{-\bar{c}(1-b)\bar{c}}{(1-b)^2 \bar{c}^2} \hat{c}_t + \frac{\bar{c} b (1-b) \bar{c}}{(1-b)^2 \bar{c}^2} \hat{c}_{t-1}$$

$$\hat{X}_t = \frac{-1}{1-b} \hat{c}_t + \frac{b}{(1-b)} \hat{c}_{t-1}$$

$$\hat{c}_t = b \hat{c}_{t-1} - (1-b) \bar{x}_t$$

$$\underbrace{- (1-\beta b)}_{1-\beta b\rho} \bar{z}_t$$

Solution for $\hat{c}_t$.

$$\boxed{\hat{c}_t = b \hat{c}_{t-1} + \underbrace{(1-b)(1-\beta b)}_{1-\beta b\rho} \bar{z}_t}$$

$$\bar{z}_t = \rho \bar{z}_{t-1} + \varepsilon_t \quad \Leftrightarrow \quad \bar{z}_t = \frac{\varepsilon_t}{1-\rho L}$$

$$L \bar{z}_t = \bar{z}_{t-1}$$

$$(1-\rho L)(1-bL) \hat{c}_t = \frac{(1-b)(1-\beta b)}{1-\beta b\rho} \varepsilon_t$$

$$(1 - \rho L)(1 - bL) \hat{c}_t$$

$$= (1 - (\rho + b)L + \rho b L^2) \hat{c}_t$$

$$= \hat{c}_t - (\rho + b)\hat{c}_{t-1} + \rho b \hat{c}_{t-2}$$

↳ solution

$$\hat{c}_t = (\rho + b)\hat{c}_{t-1} - \rho b \hat{c}_{t-2} + \frac{(1-b)(1-\rho b)}{1-\rho b} \varepsilon_t$$

AR(2)

$$\rho = b = 0$$